Fall 2024

KAIST EE412

Foundation of Big Data Analytics

Final Exam

Name: _____

Student ID: _____

I agree to comply with the School of Electrical Engineering Honor Code.

Signature: _____

This exam is open notes, closed book, and closed computer. There are 18 pages in total. You have 150 minutes (9:00 – 11:30) to complete your exam. Please use this exam sheet to write answers. You may write in English or Korean.

Read the questions carefully and focus your answers on what has been asked. You are allowed to ask the instructor/TAs for help only in understanding the questions, in case you find them not completely clear. Be concise and precise in your answers and state clearly any assumption you may have made. Good luck!

Question	Score
1	/12
2	/12
3	/20
4	/20
5	/20
6	/20
7	/20
Total	/124

1 (12 points) TRUE/FALSE

State if the following statements are true or false. Please write TRUE or FALSE in the space provided. Each problem is worth 1 point, but you will be penalized -1 point for an incorrect answer, so leave the answer blank if you don't know.

(a) Content-based recommendation is better than collaborative filtering as long as user and item profiles are given.

Answer:_____

(b) We cannot train our recommender system with the RMSE loss if the utility matrix is binary (i.e., it contains only implicit feedback).

Answer:_____

(c) It is possible that nodes with very small degrees to have high PageRank scores.

Answer:_____

(d) Betweeness of an edge is always positive.

Answer:_____

(e) Some activation functions for deep learning are not mathematically differentiable.

Answer:_____

(f) The time and space complexity of backpropagation increases quadratically with the number of layers.

Answer:_____

(g) Word2Vec creates better embeddings when the window size increases.

Answer:_____

(h) DeepWalk will still work (i.e., create meaningful embeddings) if we replace the dot product $u^\top v$ to the addition $\sum_i u_i + v_i$ for embedding similarity.

Answer:_____

(i) The number of parameters in graph neural networks is weakly proportional to the graph size (i.e., the number of edges).

Answer:_____

(j) When running a graph neural network, if the nodes in a graph do not contain initial features, it is okay to model them as learnable parameters.

Answer:_____

(k) For each new item, the complexity of reservoir sampling to deal with the new item is proportional to the size of the reservoir S.

Answer:_____

(l) Matrix sketching for streaming data cannot be more accurate than SVD.

Answer:_____

2 [12 points] Recommender Systems

(a) [6 points] *Diversity* is an important aspect to evaluate a recommender system. That is, we want each user to get a list of recommendation consisting of *diverse items* (e.g., we don't want that the list is filled with all horror movies). Suppose that we are given only the utility matrix $\mathbf{R} \in [1, 5]^{n \times m}$, where n is the number of users and m is the number of items. Describe your own way to create a recommender system that supports high diversity, with your own definition of diversity.

You'll get full credit if your answer is concise, clear, and able to convince the reader.

(b) For Homework 2, we have designed two types of recommender systems. The first one aims to predict the expected (explicit) rating r_{xi} of each user x to each item i, while the second one aims to predict if user x will interact with item i or not.

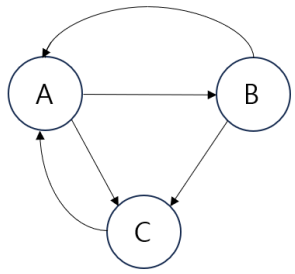
Suppose that you work for an e-commerce company like Coupang or Amazon, and have to implement your own recommender system. You are given the utility matrix consisting of explicit feedback and the timestamps as in the homework.

(i) [3 points] Describe which of the two methods above will fit better in this case, and convince the reason why. Make (reasonable) assumptions if needed.

(ii) [3 points] Introduce a scenario where the one you didn't choose in (i) is expected to be better than the one you chose in (i). Make assumptions if needed.

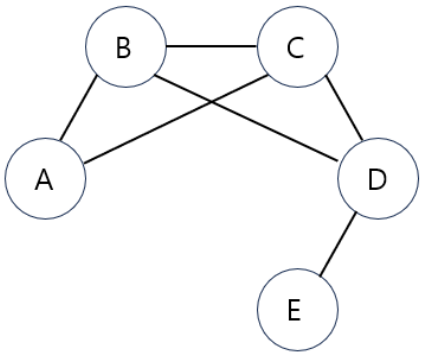
3 [20 points] Link Analysis

(a) [6 points] Consider the following web graph of three pages A, B, and C:

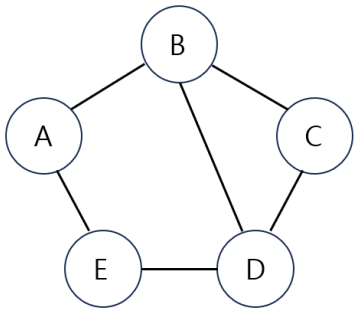


Calculate the PageRank values of all nodes (A, B, C) after one iteration. Assume that the teleport probability is $1-\beta = 0.15$, where $\beta = 0.85$. Initially, the PageRank values for all nodes are equal, i.e., $r_A = r_B = r_C = \frac{1}{3}$. Provide your answer rounded to two decimal places.

(b) [6 points] Calculate the betweenness of each edge in the graph below. Remove the edge(s) with the highest betweenness from the graph and then identify the resulting connected components. List the nodes in each component.



(c) Given the graph below, we aim to find the community assignments of nodes using the *modified* affiliation graph model. In this modified model, all clusters share the same edge probability: $p = 0.8$ within a cluster and $p = 0.2$ between clusters.



(i) [4 points] Calculate the likelihood of the observed graph for each of the following assignments of clusters C_1 and C_2. Identify which of them is better:

- **Case 1**: $C_1 = \{A, B, C\},\ C_2 = \{D, E\}$
- **Case 2**: $C_1 = \{B, C, E\},\ C_2 = \{A, D\}$

(ii) [4 points] Suppose that you are unsure if edge (D, E) really exists or not in the graph because of some noise added while collecting the data. Based on the better assignment from (i), calculate the likelihood for both cases – (D, E) exists or does not exist. Then, determine whether (D, E) is more likely to exist or not.

4 [20 points] Neural Networks and Deep Learning

(a) [6 points] The Kullback-Leibler (KL) divergence quantifies the divergence between two probability distributions P and Q, and is defined as:

$$D_{\mathrm{KL}}(P \parallel Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)},$$

where P represents the true data distribution, and Q is the model's predicted distribution parameterized by θ. The goal of maximum likelihood estimation is to find θ that maximizes the likelihood of observed data D, expressed as $p(D \mid \theta)$. Demonstrate that maximizing the likelihood is equivalent to minimizing the KL divergence $D_{\mathrm{KL}}(P \parallel Q)$, starting from the optimization objective:

$$\theta^* = \arg\min_\theta D_{\mathrm{KL}}(P \parallel Q).$$

(b) [8 points] Prove that the KL divergence is a non-negative objective function, i.e., $D_{KL}(P \parallel Q) \geq 0$, and explain the condition under which $D_{KL}(P \parallel Q) = 0$. To solve this problem, you may use a mathematical property called **Jensen's inequality**, which applies to convex or concave functions:

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)] \quad \text{if } f \text{ is a convex function.}$$
$$f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)] \quad \text{if } f \text{ is a concave function.}$$

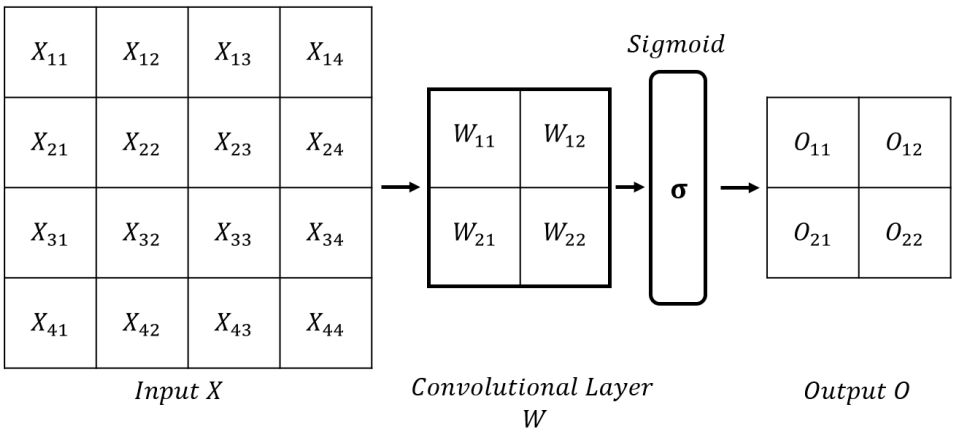
Note that a twice differentiable function $f(x)$ is **convex** if $f''(x) \geq 0$ (its second derivative is non-negative). Conversely, a function $g(x)$ is **concave** if $g''(x) \leq 0$ (its second derivative is non-positive). For instance, the natural logarithm $\log(x)$ is a concave function, which implies that $-\log(x)$ is convex.

(c) [6 points] Consider a single-layer convolutional neural network applied to a 4×4 input grid X. The convolutional layer uses a 2×2 kernel W with a stride of 2 and no padding, producing a 2×2 output matrix O. A sigmoid activation function σ is applied element-wise to the convolutional output:

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

For example, the element O_{21} of the output matrix can be expressed as:

$$O_{21} = \sigma\left(W_{11}X_{31} + W_{12}X_{32} + W_{21}X_{41} + W_{22}X_{42}\right).$$



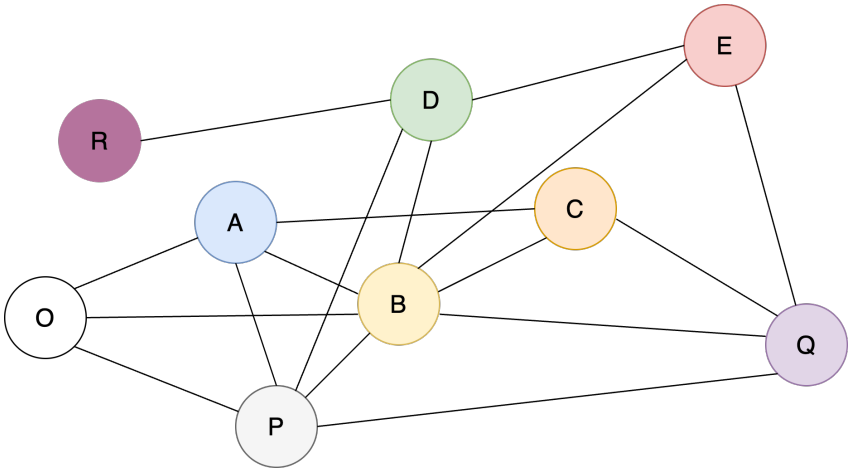
We apply the loss function L to train the network above:

$$L = \frac{1}{2}\sum_{i=1}^{2}\sum_{j=1}^{2}(y_{ij} - O_{ij})^2,$$

where y_{ij} is the target. Use the chain rule to derive the partial derivative of the loss L with respect to the input value X_{11}.

5 [20 points] Graph Representation Learning

(a) To learn node embeddings, suppose that we run a **biased random walk**, which is a part of Node2Vec, on the graph below:



Assume that we determine the values of p (return parameter) and q (in-out parameter) specifically to **the current location of the walker**. That is, if the node set is $\mathcal{V} = \{A, B, C, D, E, O, P, Q, R\}$, then the walker uses p_x and q_x at node $x \in \mathcal{V}$.

NOTE: The walker moves to the target node with the highest transition probability. In the case of a tie, the walker selects the node with the lower index based on alphabetical order (e.g., A is lower than B).

(i) [10 points] Suppose that the walker started from node O and is currently at node A. It moves as the following route:

$$A \to C \to B \to D \to E \to B \to A \to B \to D.$$

Based on this route, **tell if the values of p_x and q_x for each node** x are greater than 1, less than 1, or indeterminate. If a value cannot be determined relative to 1 (i.e., indeterminate), specify if it is greater or smaller than the other parameter (p or q), and complete the table below.

Node	$\boldsymbol{p}$	$\boldsymbol{q}$
$\boldsymbol{A}$		
$\boldsymbol{B}$		
$\boldsymbol{C}$		
$\boldsymbol{D}$		
$\boldsymbol{E}$		

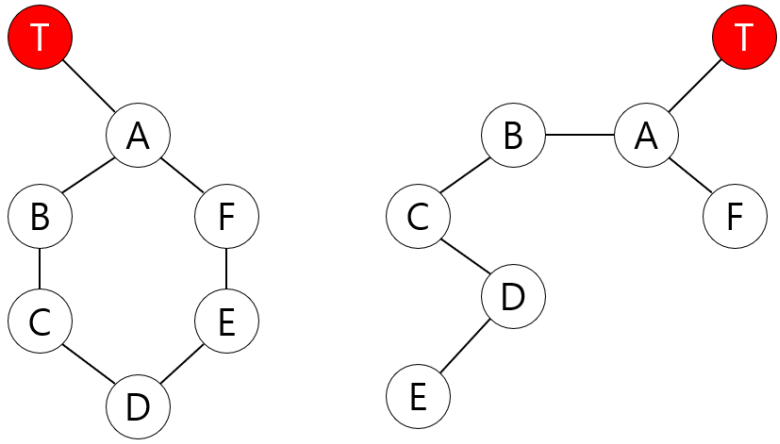
(ii) [5 points] Let's ignore the route given in (i). Suppose that the walker started at node O and is currently at node A. Assume that the walker alternates between DFS-like ($q < 1 < p$) and BFS-like walks ($1 < p < q$) (i.e., DFS $\rightarrow$ BFS $\rightarrow$ DFS $\rightarrow$ BFS) from node A. If the walker moves 5 times more to generate neighbors, determine the most likely route the walker will take.

(b) [5 points] Word2Vec is an effective method for generating word embeddings, but has several limitations as well. First, it struggles to generalize to new words not seen during the training. Second, it struggles to embed words with multiple meanings (e.g., polysemy) or to adapt to contextual changes in meaning.

What characteristics of Word2Vec lead to these limitations? Provide explanations to support your reasoning.

6 [20 points] Graph Neural Networks

(a) Consider the following two graphs below, where all nodes have 1-dimensional initial feature vectors. That is, $x_i = [1]$ for each node i in the graph.



We use a simple linear GNN with sum aggregation and without linear transformation. Specifically, at every layer t, the embedding of node v is updated as the sum over the embeddings of its neighbors (N_v) and its current embedding h_v^t:

$$h_v^{t+1} = h_v^t + \sum_{u \in N_v} h_u^t.$$

Suppose we run the GNN to compute node embeddings for the two target nodes (the red nodes with T in the figures) respectively.

(i) [4 points] How many layers of message passing are needed so that these two target nodes can be distinguished (i.e., have different GNN embeddings)?

(ii) [6 points] Let k be the number of necessary layers (i.e., the answer of (a)). Calculate h_T^k for the left and right graph, respectively. For example, if $k = 2$, h_T^2 represents the embedding of node T after two layers of message passing.

Left graph:

Right graph:

(b) In class, we have learned about LightGCN, a GNN model for recommender systems. The model updates node embeddings $\mathbf{E}$ for each layer $k \in \{0, 1, ..., K-1\}$:

$$\mathbf{E}^{(k+1)} = \tilde{\mathbf{A}}\mathbf{E}^{(k)},$$

where $\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ is the diffusion matrix, $\mathbf{A}$ is the adjacency matrix of the graph, and $\mathbf{D}$ is the degree matrix generated from $\mathbf{A}$. If you recall, LightGCN also uses multi-scale diffusion to get the final node embeddings:

$$\mathbf{E}^{(K)} = \sum_{i=0}^{K} \alpha_i \mathbf{E}^{(i)},$$

where the uniform coefficients $\alpha_i = \frac{1}{K+1}$ are typically used.

(i) [5 points] Suppose we have modified the original propagation layer into

$$\mathbf{E}^{(k+1)} = \beta\mathbf{E}^{(0)} + (1-\beta)\tilde{\mathbf{A}}E^{(k)},$$

where β is called the "teleport probability" to control the retention of starting features in the propagation.

Express the layer-K embeddings $\mathbf{E}^{(K)}$ as a function of the initial embeddings $\mathbf{E}^{(0)}$ and the diffusion matrix $\tilde{\mathbf{A}}$. Then, show that the original LightGCN can represent this modified version by properly setting α_i to a certain value.

(ii) [5 points] Let's consider a more challenging case. Assume that we have modified the propagation layer into a different one:

$$\mathbf{E}^{(k+1)} = \beta \tilde{\mathbf{A}} E^{(k)} + (1 - \beta)\mathbf{E}^{(k)},$$

In this variant, we add a residual connection that incorporates the embeddings from the previous layer directly, scaled by a weight factor. This allows for the gradual blending of previous embeddings.

Express the layer-K embeddings $\mathbf{E}^{(K)}$ as a function of the initial embeddings $\mathbf{E}^{(0)}$ and the diffusion matrix $\tilde{\mathbf{A}}$. Then, show that the original LightGCN can represent this modified version by properly setting α_i to a certain value.

7 [20 points] Mining Data Streams

In this problem, we explore extensions of the Bloom filter to support time-based decay. We define a data stream S as a sequence of N items, denoted by e_i, associated with their time-stamps t_i (note that $t_1 = 0$):

$$S = \{(e_1, t_1), (e_2, t_2), ...(e_N, t_N)\}.$$

We consider the **time-decaying Bloom filter (TBF),** which has two differences from the basic Bloom filter. First, it stores *counters* that can be incremented or decremented, instead of bits. Second, the counters decay exponentially over time. When an item arrives at time T, it adds a weight of 1 to its counters. As time progresses to $T' > T$, this weight decays to $\alpha^{(T'-T)}$, where $0 < \alpha < 1$ is the decay factor.

(a) [5 points] The goal of TBF is to focus on recent items rather than old items. Let's say we want detect recent items within the time window W while ignoring items older than W. We consider a counter "set" when its counter value exceeds a threshold ϵ. For this purpose, the threshold must satisfy:

$$\alpha^W < \epsilon < 1$$

where W is the effective time window of interest and α is the decay factor.

Explain why both (lower and upper) bounds are necessary for TBF for our purpose.

(b) [5 points] For a counter that has received hits at arrival times $\{t_1, t_2, ..., t_m\}$, write an expression for its value at current time T.

(c) Let $g_e(T)$ be the counter for item e at time T. Although this counter accumulates decayed values, it does not directly represent the decayed frequencies. For example, consider an item that appears in every time step from $t = 0$ to current time T. Its counter value would be:

$$g_e(T) = 1 + \alpha + \alpha^2 + ... + \alpha^T = \frac{1 - \alpha^{T+1}}{1 - \alpha}.$$

This value can exceed 1, making it difficult to interpret as a frequency measure. For example, $\alpha = 0.5$ gives $g_e(T) \approx 2$, and $\alpha = 0.9$ gives $g_e(T) \approx 10$ as $T \to \infty$. A proper frequency measure should be bounded in $[0, 1]$ where a value of 1 represents an item appearing at every time step, regardless of the decay rate α.

To properly represent the counter as a frequency, we introduce $f_e(T)$ as the decayed frequency value at time T:

$$f_e(T) = (1 - \alpha)h_e^{(T)} + \alpha f_e(T - 1)$$

where $\alpha \in [0, 1)$ is the decay factor and $h_e^{(t)}$ is the indicator of the hit of the item e for the exact time step t (1 if e appears at time t, 0 otherwise).

(i) [5 points] Show that this recursive definition ensures $f_e(T)$ is bounded in $[0, 1]$.

(ii) [5 points] Show that we can estimate the decayed frequency value of $g_e(t)$ as

$$f_e(t) = (1 - \alpha)g_e(t)$$

[Scratch page]